

BANARAS HINDU UNIVERSITY

B.Sc. — Semester I Mid-Semester Examination, 2024-25 (NEP)

MATHEMATICS**Paper Code: MATMJ-11 / MATMN-11 : Calculus (Mid-Semester)**

Time: 1 Hour

Maximum Marks: 20

*Instructions: Answer all questions. Each question carries equal marks (5 marks each).***Question 1.**

- (a) Define a convergent sequence of real numbers. Give an example of a convergent sequence. Show that

$$\lim_{n \rightarrow \infty} (n)^{1/n} = 1.$$

[2.5]

- (b) Define a Cauchy sequence of real numbers. Show that every Cauchy sequence of real numbers is bounded.

[2.5]

Question 2.

- (a) Define a monotonic sequence of real numbers. Give an example of a monotonic sequence of real numbers. [2]
- (b) Let $\{a_n\}$ be a sequence of real numbers such that it is monotonically increasing and it has an upper bound. Show that $\lim_{n \rightarrow \infty} a_n = \ell$ exists. Find the value of ℓ . [3]

Question 3.

- (a) Define a convergent series of real numbers. Give an example of a series of real numbers, which is not convergent and justify it. [2]
- (b) Test the convergence of the series

$$\sum_{n=1}^{\infty} \left(\sqrt{n^4 + 1} - \sqrt{n^4 - 1} \right).$$

[3]

Question 4.

- (a) Let I be an interval in $\mathbb{R}$ and let c be a limit point of I . Give ε - δ definition of limit of a function $f : I \rightarrow \mathbb{R}$ at c as x approaches to c . Give an example in support of ε - δ definition of limit of a function. [2.5]

(b) Define a function $f : (-\infty, 0) \cup (0, \infty) \rightarrow \mathbb{R}$ by

$$f(x) = \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}, \quad x \neq 0.$$

Does $\lim_{x \rightarrow 0} f(x)$ exist? Find $\lim_{x \rightarrow 0} f(x)$ if it exists.

[2.5]